

Jack Mislinski

Founder, Aevox · Metabolic-testing software, built and deployed inside clinics · MSc Applied Physiology · ex-Barclays credit trader
Brooklyn, NY · jackmis610@gmail.com · (312) 914-1773 · jackmislinski.com · github.com/jackmislinski · jackmislinski.substack.com

SUMMARY

A physiologist who builds and deploys the software himself. Founder of Aevox, a CPET and metabolic-testing platform run as one isolated deployment per clinic: five sites live on four cart vendors, from a CommonSpirit Health / EXOS performance center to a lab in Reykjavik. The same loop at every site: sit with the physiologist, run the test, ingest the cart's export as uploaded, find where the report fails, ship the fix that week, fold it into the core. MSc Applied Physiology (4.0 GPA), NYU Computer Science, ex-Barclays credit trader on a \$1B book.

EXPERIENCE

Aevox — Founder & sole engineer

2024 – Present

- **Deployment.** One isolated deployment per clinic (own Vercel project, Clerk instance, Neon Postgres in the clinic's region), provisioned by a single script with per-clinic branding, DNS, and a first-visit checklist. Five sites live; the first is the Hybl Performance Center (CommonSpirit Health / EXOS).
- **Ingest.** Accepts a lab's export as uploaded: unit-aware column mapping, a Claude-proposed mapping checked against physics (Weir energy balance, VO_2/VCO_2 ranges) before it is accepted, and a per-sheet report that states exactly why a file was refused. Handles COSMED, MGC, PNOE, and VO_2 Master exports, lactate step sheets, and O_2 -only analyzers.
- **Engine.** Analysis engine at v13 with 500+ tests: ventilatory thresholds from deterministic detectors with the model as reviewer, not decider, and nothing committed until the clinician confirms; step-protocol detection from HR and VO_2 dips; lactate thresholds by first rise and Dmax ; grade-corrected flat-equivalent paces; running economy from steady-state stages only; reports that withhold what a missing input makes false and say why on page one.
- **Field loop.** Trial users' asks become shipped features in hours: a coach's per-stage treadmill grade became grade-corrected pace zones the morning he signed up; a lab's hand-graphed lactate sheet drove the step protocol within a day; a K5 export from Iceland exposed a threshold-guardrail bug that was fixed and live before the follow-up call.
- **Go-to-market and paper.** Outbound to metabolic labs and performance centers (24% reply rate), discovery calls, pricing sheet, MSA, and order form. Set the HIPAA posture (not a business associate for cash-pay performance customers), wired the athlete consent record end to end, and rewrote the privacy and security docs to match.
- **Operations.** Run the company on scheduled Claude agents I built: they rank the day's commitments, draft outreach from a cached system prompt, read per-tenant upload and refusal metrics from every deployed instance, and queue a LinkedIn and X deployment log. I keep the veto; the agents keep the calendar.

Barclays Investment Bank — Associate, Investment Grade Credit Trading

New York, NY · 2022 – 2025

- Market-made and managed risk on a **\$1B book** of front-end IG corporate bonds: daily two-way pricing and real-time coverage of 20+ buy-side accounts; built and owned the desk's analytics stack (Python, SQL, kdb/q) for PnL attribution, volumes, market share, and risk. **Tradeweb 2023 Top 5 Performer.**

SELECTED PROJECTS

Parsel — Healthcare × AI hackathon, winning entry

2026

Privacy-preserving voice monitoring for Parkinson's: a self-referenced baseline and drift, never a diagnosis. Built the clinician dashboard and patient app from scratch (React, TypeScript; spoken prompts through an ElevenLabs proxy with browser-TTS fallback; MediaRecorder capture) on a FastAPI + Baseten backend. Live at parsel.health.

Aevox Live — real-time session monitor for an on-airway breath analyzer

2026

Typed device contract (frames, commands, events), physiological simulator, real-export replay through the same contract, WebSocket relay on a Cloudflare Worker, incremental threshold and mechanics engine, and an operator assistant in which the browser runs tools against the live device while a serverless function holds the model turn; release-branch shipping behind a whole-session test harness.

bioML — substrate utilization from wearable signals

2025

Multitask gradient-boosted model predicting VO_2 , RER, and substrate oxidation from PPG heart rate, power, CGM, and demographics. Subject-wise 5-fold CV ($n=100$): RER MAE 0.034, LT2 AUC 0.953, 38–59% better than the strongest baseline on VO_2 .

TECHNICAL SKILLS

AI / LLM: Claude API and Claude Code, tool use, adaptive thinking, cached prompts, scheduled agents, physics-checked proposals, evals.

Full-stack / infra: TypeScript, React 19, Vite, Next.js, Vercel serverless, Clerk, Neon Postgres, Cloudflare Workers, WebSockets, vitest.

ML / data: Python, PyTorch, LightGBM, scikit-learn, pandas, MLflow, SQL, kdb/q; time-series models on noisy biosignals, grouped CV.

Physiology: CPET, $\text{VO}_2/\text{VCO}_2/\text{RER}$, substrate oxidation, ventilatory and lactate thresholds, step and ramp protocols, cart validation.

EDUCATION AND ADDITIONAL

M.S. Applied Physiology, University of Colorado Colorado Springs · GPA 4.0 · 2025 – 2027. Mitochondrial bioenergetics and metabolic-cart validation (6.5% bias vs. reference); two manuscripts in preparation.

B.A. Computer Science, New York University · 2020 – 2022. **United States Military Academy, West Point**, CS major · 2018 – 2020: Dean's List every semester, Cadet First Sergeant (top 2 of ~1,100).

Ultramarathoner (Tahoe Rim Trail 100, Run Rabbit Run 50) and Pacific Crest Trail thru-hiker (2,650 mi). Spanish and Portuguese.